



Activation time-course of phonological code in silent word recognition in adult readers with and without dyslexia

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Abstract

In skilled adult readers, reading words is generally assumed to rapidly and automatically activate the phonological code. In adults with dyslexia, despite the main consensus on their phonological processing deficits, little is known about the activation time course of this code. The present study investigated this issue in both populations. Participants' accuracy and eye movements were recorded while they performed a visual lexical decision task in which phonological consistency of written words was manipulated. Readers with dyslexia were affected by phonological consistency during second fixation duration of visual word recognition suggesting a late activation of the phonological code. Regarding skilled readers, no influence of phonological consistency was found when the participants were considered a homogeneous population. However, a different pattern emerged when they were divided into two subgroups according to their phonological and semantic abilities: Those who showed better decoding than semantic skills were affected by phonological consistency at the earliest stage of visual word recognition while those who showed better semantic than decoding skills were not affected by this factor at any processing stage. Overall, the findings suggest that the presence of phonological deficits in readers with dyslexia is associated with a delayed activation of phonological representations during reading. In skilled readers, the contribution of phonology varies with their reading profile, i.e., being phonologically or semantically oriented.

Keywords Developmental dyslexia · Phonological consistency · Inter-individual differences · Eye movements · Reading profile · Adults with dyslexia

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Introduction

According to the triangle model, visual word recognition results from fast and automatic activation of orthographic, phonological, and semantic codes (Harm & Seidenberg, 2004). Numerous studies reported evidence with skilled adult readers. For example, Zeguers, Snellings, Huizenga, and van der Molen (2014) demonstrated that during word reading, skilled readers first extract the orthographic code (i.e., within the first 33 ms), and rapidly translate it into phonological code within 50 ms after word exposure (see Ferrand & Grainger, 1992, 1993, 1994 for similar results in French).

In adult readers with dyslexia, visual word recognition is impaired mainly because of deficits in phonological processing (Cavalli et al., 2016; Ramus & Szenkovits, 2008). These readers experience difficulties in tasks assessing phonological awareness such as phoneme deletion (Cavalli et al., 2016; Martin et al., 2010), verbal short-term memory (Paulesu et al., 2001; Vasic, Lohr, Steinbrink, Martin, & Wolf, 2008), and speech sound discrimination (Berent, Vaknin-Nusbaum, Balaban, & Galaburda, 2013). The specific nature of their phonological deficits is still under debate (Ramus, 2014; Szenkovits, Darma, Darcy, & Ramus, 2016): whether caused by underspecified phonological representations or by a difficult and/or delayed access to and retrieval of phonological code (Boets et al., 2013; Ramus & Szenkovits, 2008). Whatever the nature of the phonological deficit be, one consequence is impaired phonological-orthographic mapping, inducing less efficient visual word processing in this population. This interpretation is confirmed by several studies suggesting that grapheme-phoneme decoding is not as automatic in individuals with dyslexia compared to skilled readers, even in adulthood (Boets et al., 2013; Suárez-Coalla & Cuetos, 2015).

Despite the main consensus on phonological deficits in readers with dyslexia, little is known about the activation time course of phonological code during silent reading, especially in adults. The aim of the present study is to address this issue by comparing the activation time course of phonological code in adult readers with and without dyslexia when they performed a visual lexical decision task on monosyllabic French words. This question was examined through the manipulation of phonological consistency of written words (Malins et al., 2016), which refers to the number of pronunciations of a given letter string. Letter strings associated with only one pronunciation (e.g., *-uck* in *duck* or *luck*) are categorized as consistent. Letter strings associated with more than one pronunciation (e.g., *-ough* in *tough* and *plough*) are categorized as inconsistent.

In skilled readers, behavioral studies showed a phonological consistency effect: reading phonological inconsistent words induce slower latencies than reading consistent ones in visual lexical decision, naming, and silent reading tasks (Graves, Desai, Humphries, Seidenberg, & Binder, 2010; Lacruz & Folk, 2004; Malins et al., 2016). This effect has been interpreted as reflecting an activation of competing pronunciations making the recognition of inconsistent words more difficult (Graves, Desai, Humphries, Seidenberg, & Binder, 2010; Malins et al., 2016). Moreover, using naming tasks, several studies showed increased activation in several brain regions during the processing of inconsistent words, thus suggesting that more resources are needed to process phonological inconsistency (Graves, Desai, Humphries, Seidenberg, & Binder, 2010; Malins et al., 2016). Such pattern of activations has also been found with briefly presented targets (Peng et al., 2004), suggesting an automatic reliance on phonological information during visual word recognition in skilled adult readers. However, the robustness of this effect remains debatable, especially in silent reading tasks—such as lexical decision—

that do not explicitly require an activation of phonological code (Lacruz & Folk, 2004; Taft & van Graan, 1998; Ziegler, Petrova, & Ferrand, 2008).

In individuals with dyslexia, very few studies have examined the influence of phonological consistency during silent word recognition (Bosman, Vonk, & van Zwam, 2006; Davies & Weekes, 2005) and, to our knowledge, only one investigated this question in adults (Kunert & Scheepers, 2014). Kunert and Scheepers (2014) manipulated words' phonological consistency and recorded eye movements while participants performed an adapted version of the lexical decision task involving a forced choice between two alternatives. In this study, a word and a pseudoword were simultaneously displayed on the screen for 3 s. Participants were required to fixate the real word as soon as they recognized it. This experimental design was employed to avoid the speed-accuracy trade-off induced by the classical lexical decision paradigm. Overall, readers with dyslexia were slower than skilled readers. Although both groups were slower to recognize inconsistent compared to consistent targets, only the accuracy scores of readers with dyslexia were further affected by the target's phonological consistency. Based on these findings, the authors concluded that adults with dyslexia demonstrated deficits in both general processing speed and phonological processing. However, the authors did not investigate the influence of phonological consistency on more fine-grained markers like first and second fixations. Therefore, no information on the precise time course of phonological code activation could be inferred.

The present study

The present study specifically focused on the contribution of phonological codes to reading in university students with dyslexia. We argued that, despite their good level of vocabulary and academic success (Cavalli et al., 2016), this high-functioning dyslexic population might still show subtle phonological deficits even in a relatively elementary reading task. To this aim, participants performed a lexical decision task while we manipulated the phonological consistency of written words and recorded participants' eye movements. By using eye-tracking measures, we could compare the eye gaze patterns of readers with and without dyslexia and investigate the activation time course of phonological code during written word recognition at a more fine-grained level than allowed by traditional behavioral measures (i.e., reaction times and accuracy). Previous studies showed that participants often make multiple fixations even during isolated word recognition, which make the examination of eye gaze pattern feasible (Miwa & Dijkstra, 2017). In addition, to further ensure that participants' eyes would not remain static, participants were required to respond by eyes rather than by hand as in the classical design of word recognition tasks (the specificities of the experimental design are further developed in the method section). In order to cover a wide range of processing stages, several eye movement measures were considered. Gaze duration (i.e., the sum of all fixations falling on a word before leaving it for the first time) and single fixation duration (i.e., the duration of the first fixation on a word when only one was made during the first reading pass) are both considered relatively coarse measures reflecting cumulative processing efforts (Leininger, 2014; Rahmanian & Kuperman, 2018; Rayner, Pollatsek, & Binder, 1998) and will therefore be taken as global measures indexing late processing stages of single word recognition. First fixation duration (i.e., the duration of the first fixation on a word when more than one was made on it during the first pass) and second fixation duration (i.e., the duration of the second fixation falling on a word during the first pass) are taken as more fine-grained measures reflecting initial (Mousikou & Schroeder, 2019; Sereno, Hand, Shahid, Yao, &

O'Donnell, 2018) and subsequent (Colé, Pynte, & Andriamamonjy, 2003; Rahmadian & Kuperman, 2018; Reichle, Rayner, & Pollatsek, 2003) processing stages, respectively. Finally, refixation probability (i.e., the probability that a word will be fixated more than once) can be taken as an indication of the proportion of words read by the direct (orth-sem) and phonologically mediated (orth-phon-sem) pathways (Hawelka, Gagl, & Wimmer, 2010): Words fixated only once are assumed to be recognized via the direct pathway while words fixated more than once would be recognized via the phonologically mediated one.

Based on previous studies, we expected readers with dyslexia to exhibit longer fixations than skilled readers on every measure (i.e., gaze duration, single, first and second fixations) and to make more refixations (Hawelka, Gagl, & Wimmer, 2010). Regarding the influence of the phonological code, as words fixated more than once are assumed to be recognized via phonologically mediated pathway, their recognition would be more sensitive to the manipulation of phonological consistency of the rime. For these words, if phonological code is readily activated during early processing stages of visual word recognition, as it could be expected in skilled readers, consistency effect should affect the early measures of visual inspection (e.g., first fixation duration, see Sparrow & Miellet, 2002). However, the presence of phonological deficits in readers with dyslexia may interfere with this process. In this population, one could expect the influence of phonological code to be delayed (e.g., to occur only during second fixation duration) or absent. Given that words fixated only once are assumed to be processed by means of the direct pathway, phonological consistency of word's rime should not affect single fixation durations. Finally, as gaze duration is a more global and coarse measure encompassing both direct and phonologically mediated pathways, the influence of phonological consistency on this measure may be unstable, as previously reported for behavioral measures (Lacruz & Folk, 2004; Ziegler, Petrova, & Ferrand, 2008).

Method

Participants

The experiment was conducted in accordance with the Declaration of Helsinki and with the understanding and written consent of each participant. The project was approved by the local ethics committee (Aix-Marseille University, Marseille, France). We recruited 30 university students with dyslexia (15 women) and 30 gender-matched skilled readers. All were French native speakers from Aix-Marseille University, were right-handers, had a normal or corrected to normal vision, and had non-verbal IQ (Raven's Matrices: Raven, Raven, & Court, 1998) within the normal range (above the 75th percentile). None of them reported neither neurological nor psychiatric disorder. All participants with dyslexia reported major difficulties in learning to read during childhood and had received a formal diagnosis of dyslexia.

As reported in Table 1, the two groups were also matched on chronological age (mean, 21.45 years old), educational level (mean, 1.85 years of higher education), vocabulary knowledge (the EVIP scale, Dunn, Theriault-Whalen, & Dunn, 1993), and non-verbal IQ. They were also matched on field of study (humanities and social science, psychology, biology, neuroscience, and mathematics).

Reading efficiency was assessed with a French reading test standardized for adults (*l'Alouette*, Cavalli et al., 2018), taking both accuracy and speed into account. Efficiency of

Table 1 Cognitive profile of readers with and without dyslexia. Standard deviations are reported in parentheses. Efficiency scores were obtained by dividing accuracy scores with the averaged speed (in ms) for each participant. *T* values were obtained from paired student *t* tests comparing the two groups of participants (***p* < 0.001; ns: *p* > 0.10)

	Readers with dyslexia	Skilled readers	<i>t</i> values	
Chronological age	21.6 (2.3)	21.3 (2.3)	0.356	ns
Years of higher education	1.8 (1.5)	1.9 (1.4)	−0.108	ns
Raven's matrices (max = 58)	45.1 (6.1)	47.1 (5.5)	−1.041	ns
Pretests				
Vocabulary score (<i>EVIP</i> , max = 51)	39.8 (4)	40.4 (4.1)	−0.415	ns
Reading efficiency score (<i>Alouette</i>)	332.6 (65.6)	483.6 (55.1)	−7.924	***
Irregular word reading (efficiency)	1.2 (0.2)	1.4 (0.2)	−4.877	***
Pseudoword reading (efficiency)	0.8 (0.2)	1.2 (0.2)	−6.335	***
Phonemic awareness (efficiency)	0.4 (0.1)	0.7 (0.1)	−9.721	***
Verbal STM (span)	4.6 (0.8)	5.6 (0.6)	−4.523	***

the direct and phonologically mediated reading pathways (Pritchard, Coltheart, Palethorpe, & Castles, 2012) was assessed with irregular words and pseudoword reading tasks from EVALEC (Sprenger-Charolles, Colé, Béchennec, & Kipffer-Piquard, 2005). Phonological skills involved in reading were assessed through EVALEC's initial phoneme deletion and verbal short-term memory tasks. The two groups differed significantly in reading and phonological skills (Table 1).

Stimuli

Twenty pairs of phonologically consistent (e.g., *cloche* where *–oche* can only be pronounced /ɔʃ/ in French) and inconsistent (e.g., *clef* where *–ef* can be pronounced /e/ or /ɜf/) monosyllabic words were constructed (Ziegler, Jacobs, & Stone, 1996). Within each pair, words were matched on gender, first letter, and first phoneme. As can be seen in Table 2, they were also matched on token and type bigram log frequency, spoken and written log frequency, and orthographic neighborhood (*lexique.org*, New, Pallier, Ferrand, & Matos, 2001). However, due to material constraints, the words' pairs could not be matched on number of phonemes and letters, and on orthographic and phonological uniqueness points. This will be considered when analyzing the results. A list of the words used in the present study is available in the Appendix Table 6.

Table 2 Psycholinguistic characteristics of consistent and inconsistent words. Standard deviations are reported in parentheses. *T* values were obtained from paired student *t* tests comparing the two words' categories on each variable (**p* < 0.05; ***p* < 0.01; ns: *p* > 0.10)

	Consistent words	Inconsistent words	<i>t</i> values	
Token bigram frequency (log)	3.72 (0.17)	3.72 (0.24)	0.067	ns
Type bigram frequency (log)	4.22 (0.23)	4.21 (0.32)	0.128	ns
Spoken frequency (log)	1.37 (0.54)	1.37 (0.88)	0.01	ns
Written frequency (log)	1.54 (0.47)	1.42 (0.83)	0.539	ns
Number of letters	5.4 (0.8)	4.3 (1.1)	3.488	**
Number of phonemes	3.7 (0.5)	3.2 (0.6)	2.939	*
Orthographic uniqueness point	5.4 (0.8)	4.3 (1.1)	3.317	**
Phonological uniqueness point	3.7 (0.5)	3.2 (0.6)	2.939	*

For the purpose of the lexical decision task, 40 pseudowords were created by replacing initial letters of the critical words.

Procedure

Participants were tested individually in a dimly lit room. They were sitting at 60 cm from the screen, with a chinrest and a forehead rest. Subjects' gaze location was recorded at 1000 Hz with an EyeLink 1000 eye tracker (Eyelink 1000 Tower Mount distributed by SR Research Ltd., Mississauga, Ontario, Canada). Before each session, nine-point gaze calibration and validation were performed and repeated until the validation error was smaller than 1° on average and than 1.5° for the worst point. Ocular data were used online and recorded with the Data Viewer software (SR Research Ltd.). Only the position of the dominant eye was recorded. As several previous studies showed that there is no clear relationship between eye dominance and reading abilities (Fagard, Monzalvo-Lopez, & Mamassian, 2008; Newman, Wadsworth, Archer, & Hockly, 1985), this variable was not taken into account in the subsequent analyses.

Stimuli were displayed at the center of a 21-in. CRT monitor (ViewSonic P227f, refresh rate = 85 Hz, resolution = 1600×1200 pixels), in black (luminance, 0.3 cd/m^2) on a light gray background (luminance, 60 cd/m^2) in a *Courier New* font ($x\text{-height} = 0.25^\circ$). Participants were asked to avoid any movement during the experiment. They were instructed to decide as fast and accurately as possible if the stimuli were real words. Half of the participants had to provide a "yes" response by looking at the point in the top right of the screen and a "no" response by looking at the point in the bottom right (● in Fig. 1). The other half of participants received the reversed instructions.

Each trial started with a fixation screen containing two vertical bars at the center (Fig. 1, panel a). Participants were required to fixate the area between these bars (fixation area, an invisible 1° window). Once ready, they pressed a button on the response box while keeping the eyes on the fixation area. A stimulus was immediately displayed at the center of the screen and remained visible until participants provided a response. Participants were instructed to blink between trials.

Stimuli were divided into 2 blocks of 40 items (20 words, 20 pseudowords). A new pseudo-randomized list of stimuli was generated for each pair of readers with and without dyslexia.

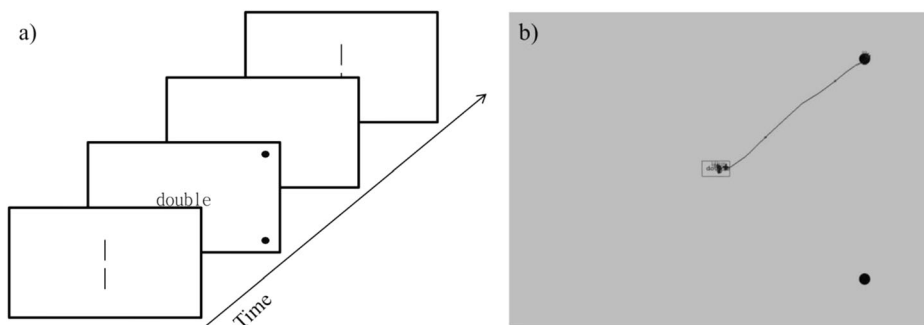


Fig. 1 Panel **a** shows an illustration of the timed-sequence of a trial. Panel **b** shows a screenshot of a trial presentation. The area within the gray box (invisible to the participants) corresponds to the interest area. The gray line illustrates the fixations and a trajectory of a participant's eyes when reading the stimulus and giving a response

The session started with a practice session containing 26 additional stimuli (13 words, 13 pseudowords).

Data analysis

Data of two individuals with dyslexia (and therefore of their paired skilled readers) were excluded from the analysis due to long fixation duration (deviating more than 2.5 std. of the mean of the group) and atypical ocular patterns. Two inconsistent words (and their associated consistent words) were further discarded due to unusually high error rates ($\geq 66\%$, corresponding to $p < 0.05$ according to a binomial test).

To compare the performances and the activation time-course of phonological code in readers with and without dyslexia, six dependent variables were considered: response accuracy, gaze duration, refixation probability, single, first and second fixation durations. The first four variables provided information on global processes of word recognition (Rahmanian & Kuperman, 2018). The two latter provided fine-grained measures of the activation time-course of phonological code (Colé, Pynte, & Andriamamonjy, 2003; Sparrow & Mielle, 2002).

Incorrect responses, fixations falling outside the target's interest area (see Fig. 1), and trials with blinks or with fixation durations longer or shorter than the mean ± 2.5 std. computed on each subject and in each condition were excluded (see Townsend, 2018 for a similar procedure). Statistical analyses were performed using *R* (Version 3.5.0) and the *lme4* package (Bates, Mächler, Bolker, & Walker, 2015). Fixation durations were analyzed through linear mixed effects models (LME, Baayen, 2008), while accuracy scores and refixation probability (i.e., binary data) were analyzed through generalized linear mixed effects models (gLME). To select the model that best explains the data and to control for the unmatched psycholinguistic characteristics, we added these unmatched factors as covariates (i.e., number of letters, of phonemes, orthographic uniqueness point, and phonological uniqueness point). Then, we compared each of the combinations we could make with these factors as covariate to the simplest model through the *anova* function from *lme4* package in order to select the best adapted model as indicated by the lowest Akaike information criterion (Matuschek, Kliegl, Vasishth, Baayen, & Bates, 2017). As the analysis relies on a factorial design, contrasts were assessed with the method of “estimated marginal means” (Searle, Speed, & Milliken, 1980) using *emmeans* (Lenth, 2016) and *multcomp* (Hothorn, Hornik, & Zeileis, 2006) packages. This method estimates the marginal means predicted by a given model and proceeds with estimation of various types of contrasts (with Tukey adjustments).

Results

Eye gaze pattern during word reading

For four out of the six dependent variables (i.e., refixation probability, single, first and second fixations' durations), adding random slopes for subjects and items and the number of letters as covariate significantly improved the model to a greater extent than any other covariate or combination of covariates. To keep coherence between the analyses, we created the same LME/gLME model for all dependent variables. The model contains two fixed effects (“group”: readers with vs. without dyslexia; “consistency”: consistent vs. inconsistent words), their interaction, the covariate “number of letters,” and random intercepts with random slopes for

subjects and items. For the two variables with which adding random slopes for subjects and items as well as the number of letters as covariate did not improve the model (i.e., accuracy and gaze duration), we ensured that it did not significantly decrease the model fit. We first investigated the eye gaze pattern of readers with and without dyslexia, independently of the manipulation of phonological consistency. The mean values of each of the six dependent variables as well as the F ratios of the main effects of group are reported in Table 3. Unsurprisingly, readers with dyslexia made longer fixations than skilled readers on all fixation duration measures (i.e., gaze duration, single, first and second fixation duration). Although they also tended to be more accurate than skilled readers, the difference did not reach significance. Interestingly, skilled readers mostly made only one fixation on the target word (for 82.1% of trials) while readers with dyslexia fixated only once less often (64.3% of trials).

Phonological code activation

Mean accuracy scores and refixation probabilities for each group and condition are reported in Fig. 2. Analysis on accuracy scores showed no main effect of consistency [$F < 1$]. The interaction between group and consistency was significant [$F(1, \text{inf}) = 4.32$, $p < 0.04$]. Post hoc comparisons revealed that simple effects of consistency were not significant in either group (readers with dyslexia: estimates = 0.65, standard deviation = 0.43, z ratio = 1.535, $p = 0.13$; skilled readers: estimates = -0.20, standard deviation = 0.40, z ratio = -0.512, $p = 0.61$), although the advantage of consistent over inconsistent words seems somewhat larger in the dyslexic group. Analysis on refixation probability revealed neither main effect of consistency nor interaction between group and consistency [both F s < 1].

Mean fixation durations for each group and condition are reported in Fig. 3. The analysis on gaze duration and single fixation duration showed neither main effect of consistency nor interaction between group and consistency for any of these measures [all F s < 1].

The analysis performed on first fixation duration revealed no significant main effect of consistency [$F < 1$]. The interaction between group and consistency failed to reach significance [$F(1, 27) = 2.99$, $p = 0.10$]. Finally, the analysis of second fixation duration showed no main effect of consistency [$F < 1$] but a significant interaction between group and consistency [$F(1, 20) = 5.50$, $p < 0.03$]. Post hoc comparisons showed that readers with dyslexia made longer fixations on inconsistent than on consistent words (estimates = -44.16, standard deviation = 20.47, t ratio = -2.157, $p = 0.04$), while skilled readers showed no difference between the two conditions (estimates = 27.56, standard deviation = 25.45, t ratio = 1.083, $p = 0.29$).

Table 3 Mean performance obtained in readers with and without dyslexia for the six dependent variables. Standard deviations are reported in parentheses. F ratios and p values correspond to the main effect of group and were obtained from LME/gLME models (* $p < 0.05$; *** $p < 0.001$; ns: $p > 0.10$)

Measure	Skilled readers	Readers with dyslexia	Main effect of group	
Gaze duration (ms)	428 (109)	593 (186)	$F(1, 62) = 53.87$	***
Accuracy (%)	87.7 (32.9)	92.0 (27.1)	$F(1, \text{Inf}) = 3.28$	ns
Single fixation duration (ms)	432 (112)	557 (190)	$F(1, 60) = 26.92$	***
First fixation duration (ms)	180 (96)	234 (153)	$F(1, 39) = 6.63$	*
Second fixation duration (ms)	229 (109)	326 (173)	$F(1, 42) = 18.54$	***
Refixation probability (%)	17.9 (38.4)	35.7 (47.9)	$F(1, \text{Inf}) = 13.78$	***

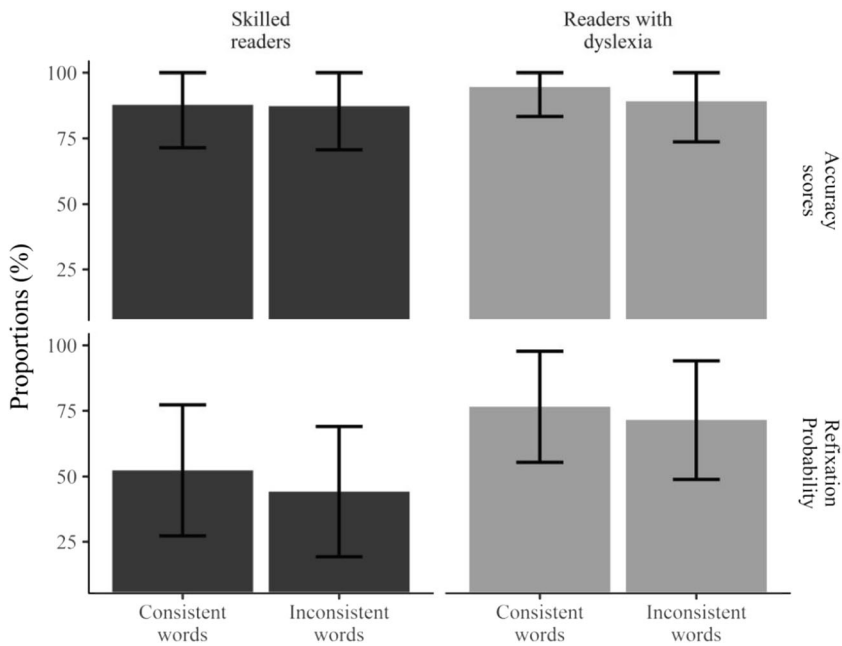


Fig. 2 Mean proportions of correct responses (top panel) and refixation probability (bottom panel) on consistent and inconsistent words in readers with (light gray) and without (dark gray) dyslexia. Error bars represent standard deviations

Several interpretations can be put forward to explain the absence of the phonological consistency effect in skilled readers. One comes from the fact that the words used in the experiment were very short and frequent (see Table 2) and were mostly recognized in only one fixation (82.1% of trials). Since they were most likely recognized via the direct pathway (Difalcis, Ferreres, Osiadacz, & Abusamra, 2018), the impact of the manipulation of

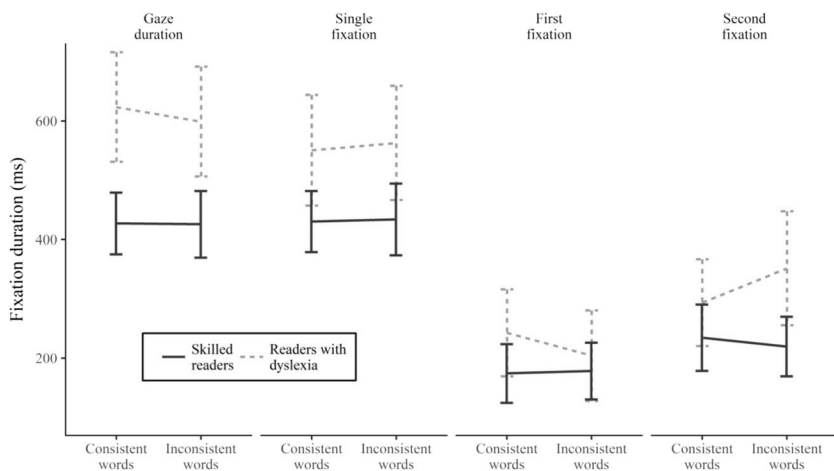


Fig. 3 Mean durations of fixations on consistent and inconsistent words in readers with (dashed light gray lines) and without (dark lines) dyslexia for gaze duration, single, first, and second fixations. Error bars represent standard deviations

phonological consistency on a sublexical unit might have been reduced. Previous behavioral studies indeed showed that this kind of manipulation had stronger effect for low-frequency words than for high-frequency words (Seidenberg, Waters, Barnes, & Tanenhaus, 1984).

Another interpretation, more related to the characteristics of participants, comes from recent studies manipulating the phonological consistency of written words in naming tasks. These studies suggest that the influence of phonological consistency in skilled readers may depend on individuals' reading profile, that is, whether they rely more on their phonological or semantic skills when they recognize written words (Hoffman, Lambon Ralph, & Woollams, 2015; Woollams, Lambon Ralph, Madrid, & Patterson, 2016). Using a naming task, Woollams, Lambon Ralph, Madrid, and Patterson (2016) investigated this question by manipulating semantic (i.e., imageability) and phonological (i.e., consistency) properties of written words that they presented to skilled readers with different reading profiles. The authors showed that readers with high semantic reliance exhibited a larger imageability effect than readers with low semantic reliance when they processed inconsistent words, but not when they processed consistent words. Furthermore, when using low imageability words, readers with high semantic reliance were more affected by phonological consistency than readers with low semantic reliance. They interpreted these results as suggesting that skilled readers with relatively higher semantic but lower phonological skills compared to their peers (henceforth, "semantic readers") seemed to rely more on the connections between the orthographic and semantic codes when recognizing words, and had less well-developed phonological representations. Although the causal link remains to be investigated, they speculated that this mild disadvantage in phonological skills might have increased their reliance on semantic activation to cope with the processing load induced by phonological inconsistency (Woollams & Patterson, 2012). However, those who showed the opposite profile (henceforth, "phonological readers") seemed to rely more on the connections between orthography and phonology regardless of the imageability degree of the stimuli (see also Hoffman, Lambon Ralph, & Woollams, 2015).

As mentioned above, the literature on skilled readers suggests a negative correlation between semantic and phonological abilities (Hoffman, Lambon Ralph, & Woollams, 2015; Woollams, Lambon Ralph, Madrid, & Patterson, 2016; Woollams & Patterson, 2012). In the present sample of skilled readers, we indeed found such correlation between the raw scores of vocabulary knowledge (which highly contributes to proficient semantic reading skills, according to Nalóm, Soares, & Cármió, 2015) and the efficiency scores of pseudoword reading ($r = -0.390$, $p = 0.04$; see Fig. 4, right panel). The same correlation was not significant in readers with dyslexia ($r = -0.004$, $p = 0.98$; see Fig. 4, left panel).

Thus, we decided to conduct post hoc analyses taking into account the inter-individual differences among skilled readers and examined whether the activation time-course of the phonological code depended on their reading profile. We expected "phonological readers" to exhibit a stronger and probably earlier influence of phonological consistency than "semantic readers" who are more likely to bypass the phonological code activation during silent reading.

Moreover, despite the absence of a significant correlation between vocabulary knowledge and phonological decoding skills in individuals with dyslexia, we also conducted post hoc analyses examining the inter-individual differences in this group of participants. These additional analyses were motivated by two arguments. First, the emerging literature on protective and compensatory mechanisms in readers with dyslexia (Cavalli et al., 2017; Farris, Ring, Black, Lyon, & Odegard, 2016; Haft, Myers, & Hoeft, 2016; Moojen et al., 2020). In particular, several studies reported that some individuals with dyslexia show a

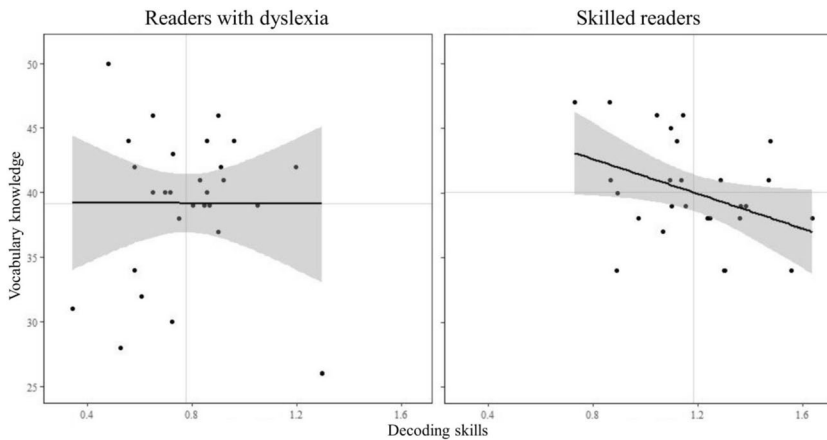


Fig. 4 Performances of individual subjects on vocabulary knowledge (y-axis, raw scores) and phonological decoding skills (x-axis, efficiency scores of pseudoword reading) for readers with dyslexia (left panel) and skilled readers (right panel). The black lines represent the correlations between the two variables and the gray areas represent standard deviations. The light gray lines represent the mean performance of each group on each variable

relatively good comprehension level despite poor decoding skills, which highlights a discrepancy in these two types of skills in readers with dyslexia (Aboud, Bailey, Petrill, & Cutting, 2016; Cutting, Bailey, Barquero, & Aboud, 2016; Patael et al., 2018; Welcome, Leonard, & Chiarello, 2010). Second, the fact that linear correlations are not always the best way to investigate relations between different skills, especially when there is a strong heterogeneity among individuals. Similarly to the skilled readers group, we hypothesized that readers with dyslexia who showed better semantic skills (hereafter, *sem+* dyslexics) could be less influenced by phonological consistency than those who showed better phonological skills (hereafter, *sem-* dyslexics). However, in the latter group, the phonological consistency effect was expected to occur only on the second fixation (which reflects a late processing stage) given the lack of strong and efficient connections between the phonological and orthographic codes in this population.

Post hoc analyses: exploring inter-individual differences in the contribution of phonology to reading among skilled readers

Based on Woollams, Lambon Ralph, Madrid, and Patterson's (2016) assumptions and on the negative correlation we observed, we classified our skilled readers as phonological or semantic readers according to their performance on pseudoword naming and vocabulary tasks. For each of these two variables, we calculated the median of the group and subtracted it from the subjects' scores. Following previous studies (Hoffman, Lambon Ralph, & Woollams, 2015; Woollams, Lambon Ralph, Madrid, & Patterson, 2016), we ranked each skilled reader from 1 to 28 (from the highest to the lowest value). To take into account both variables, we subtracted the rank position obtained in the pseudoword naming task from the one obtained in the vocabulary task. Phonological readers obtained positive scores (i.e., higher decoding skills than vocabulary knowledge), while semantic readers obtained negative scores (i.e., higher vocabulary knowledge than decoding skills). As described in Table 4, the comparison of the cognitive profiles of the two subgroups indeed showed that the semantic and phonological readers performed differently in tasks involving vocabulary, phonological decoding, and

phonological STM while showing similar reading performance on the *Alouette* Test which confirmed that, despite their different reading profile, the two subgroups were matched on overall reading efficiency.

The role of phonology in reading was examined through LME/gLME models. We applied the same selection procedure as in the first analyses to select the model that best fits the data. For three out of the six dependent variables (i.e., gaze duration, accuracy, and single fixation duration), adding random slopes for subjects and items and the covariate “orthographic uniqueness point” significantly improved the model. Therefore, we created the same LME/gLME model for the six dependent variables. The model included two fixed effects (“subgroup”: phonological vs. semantic readers; “consistency”: consistent vs. inconsistent words), their interaction, the covariate “orthographic uniqueness point,” and random intercepts with random slopes for subjects and items. Again, adding random slopes for subjects and items as well as the orthographic uniqueness point as covariate did not systematically improve the model (i.e., refixation probability and first and second fixation durations). In those specific cases, we ensured that it did not significantly decrease the model fit. To ensure comparability with the comparison between readers with and without dyslexia, we also conducted the analyses using the same model as in the initial analyses, i.e., using “number of letters” instead of “orthographic uniqueness point” as covariate. The results are available in the Appendix Table 7 and were similar to those obtained with the model including “orthographic uniqueness point” as covariate.

Mean accuracy scores and refixation probabilities for each subgroup and each condition are reported in Fig. 5. Analysis on accuracy scores showed no main effect of consistency and no interaction between subgroup and consistency [both F s < 1]. Analyses on refixation probability did not reveal either main effect of consistency or interaction between subgroup and consistency [both F s < 1].

Mean fixation durations for each subgroup and condition are reported in Fig. 6. Analysis on gaze duration revealed neither main effect of consistency [$F(1,27) = 1.37$, $p = 0.25$], nor interaction between subgroup and consistency [$F(1,23) = 1.18$, $p = 0.29$]. The same pattern of results was observed when analyzing single fixation duration: no main effect of consistency [$F < 1$], and no interaction between subgroup and consistency [$F(1,20) = 1.71$, $p = 0.21$].

We then examined the first and second fixation durations. However, since words that were fixated more than once represent only 17.9% of the correctly recognized words on average (25.9% in phonological readers, 10.1% in semantic readers), these results must be taken with caution. The analysis of first fixation duration showed no main effect of consistency [$F < 1$].

Table 4 Cognitive profile of “phonological” and “semantic” skilled readers. Standard deviations are reported in parentheses. Efficiency scores were obtained by dividing accuracy scores by the averaged speed (in ms) for each participant. T values were obtained from paired student t tests comparing the two subgroups of participants (*** $p < 0.001$; * $p \leq 0.05$; ns: $p > 0.10$)

	Phonological readers	Semantic readers	t values	
Raven’s matrices (max = 58)	49.6 (4.5)	49.7 (5.4)	− 0.044	ns
Reading efficiency score (Alouette)	496.2 (30.5)	474.2 (55.8)	0.192	ns
Pseudoword reading (efficiency)	1.3 (0.2)	1.0 (0.2)	4.983	***
Vocabulary score (EVIP, max = 51)	37.6 (2.9)	42.5 (3.2)	− 4.916	***
Irregular word reading (efficiency)	1.5 (0.2)	1.3 (0.2)	4.490	***
Phonemic awareness (efficiency)	0.8 (0.1)	0.7 (0.2)	1.501	ns
Verbal STM (span)	5.8 (0.4)	5.4 (0.7)	2.482	*

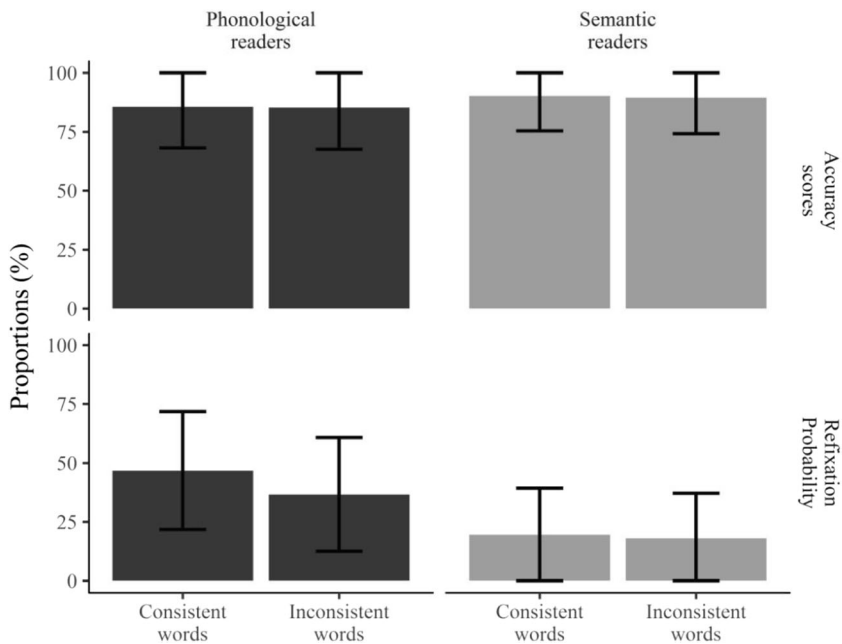


Fig. 5 Mean proportions of correct responses (top panel) and refixation probability (bottom panel) on consistent and inconsistent words in phonological (dark gray) and semantic readers (light gray). Error bars represent standard deviations

However, the interaction between subgroup and consistency was significant [$F(1,9) = 6.09$, $p < 0.04$]. Interestingly, post hoc comparisons indicated that phonological readers were affected by phonological consistency: they tended to make longer fixations on inconsistent than on consistent words (estimates = -42.29 , standard deviation = 21.03 , t ratio = -2.011 , $p = 0.06$) while semantic readers were not influenced by this factor (estimates = 41.49 , standard

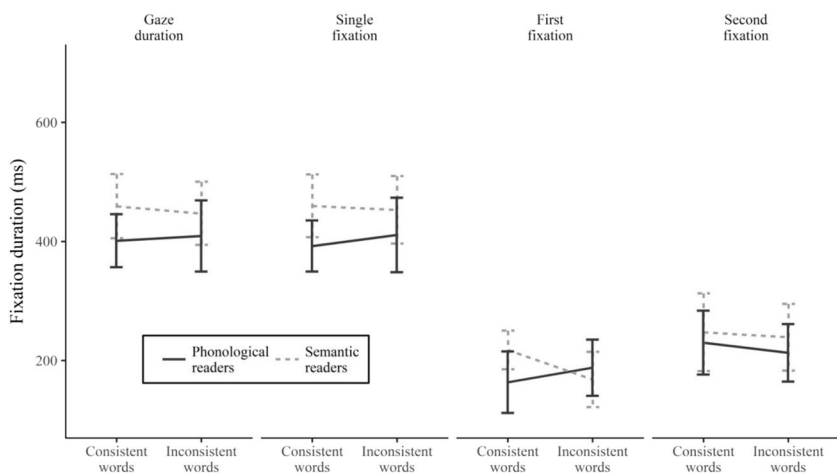


Fig. 6 Mean fixation durations on consistent and inconsistent words in phonological (dark lines) and semantic readers (dashed light gray lines) for gaze duration, single, first, and second fixations. Error bars represent standard deviations

deviation = 29.22, t ratio = 1.420, $p = 0.18$). For second fixation duration, the analyses did not reveal either main effect of consistency or interaction between subgroup and consistency [both F s < 1].

Post hoc analyses: exploring inter-individual differences in the contribution of phonology to reading among readers with dyslexia

The emerging literature on protective and compensatory mechanisms in readers with dyslexia suggests that strong morphological awareness (Cavalli et al., 2017; Haft, Myers, & Hoeft, 2016; Law, Wouters, & Ghesquière, 2015) and/or strong vocabulary skills (Cavalli et al., 2016; Haft, Myers, & Hoeft, 2016; Rose & Rouhani, 2012) have protective effects against poor reading outcomes. Moreover, it has been reported that some readers with dyslexia exhibited a discrepancy between decoding and comprehension skills (Aboud, Bailey, Petrill, & Cutting, 2016; Cutting, Bailey, Barquero, & Aboud, 2016; Patael et al., 2018; Welcome, Leonard, & Chiarello, 2010). Therefore, despite the absence of linear correlation between vocabulary knowledge and phonological skills in our sample of readers with dyslexia ($r = -0.004$, $p = 0.98$; Fig. 4, left panel), it was still worth examining whether the contribution of phonology to reading was modulated by inter-individual differences regarding these two abilities. Readers with dyslexia were divided into sem+ and sem- dyslexics following the same procedure applied on skilled readers. As shown in Table 5, the comparison of the cognitive profile of the two subgroups showed that they only differed on pseudoword reading efficiency and vocabulary scores, which are the two variables that we used to separate them. This observation suggests that the two subgroups were matched on overall reading efficiency and phonological skills.

Again, the role of phonology in reading was examined through LME/gLME models and we applied the same selection procedure as described above. For four out of the six dependent variables (i.e., durations of single fixation, first and second fixations, and accuracy scores), adding random slopes for subjects and items as well as the covariate “number of letters” significantly improved the model. For the two other variables (i.e., gaze duration and second fixation duration), this did not significantly decrease the model fit. We therefore created the same LME/gLME model for the six dependent variables. The model included two fixed effects (“subgroup”: sem- vs. sem+ dyslexics; “consistency”: consistent vs. inconsistent words), their interaction, the covariate

Table 5 Cognitive profile of “sem-” and “sem+” dyslexics. Standard deviations are reported in parentheses. Efficiency scores were obtained by dividing accuracy scores by the averaged speed (in ms) for each participant. T values were obtained from paired student t tests comparing the two subgroups of participants ($*p \leq 0.05$; ns: $p > 0.08$)

	Sem- dyslexics	Sem+ dyslexics	t values	
Raven's matrices (max = 58)	48.71 (48.7)	48.07 (48.1)	0.372	ns
Reading efficiency score (Alouette)	365.91 (365.9)	338.97 (339)	1.136	ns
Pseudoword reading (efficiency)	0.88 (0.9)	0.68 (0.7)	2.728	*
Vocabulary score (EVIP, max = 51)	36.57 (36.6)	41.79 (41.8)	-2.714	*
Irregular word reading (efficiency)	1.23 (1.2)	1.1 (1.1)	1.803	ns
Phonemic awareness (efficiency)	0.43 (0.4)	0.42 (0.4)	0.187	ns
Verbal STM (span)	4.5 (4.5)	4.79 (4.8)	-0.968	ns

“number of letters,” and random intercepts with random slopes for subjects and items. To ensure comparability with the model applied on the data on skilled readers, we also conducted the analyses using “orthographic uniqueness point” instead of “number of letters” as covariate. The results are available in Appendix Table 8 and were similar to those obtained with the model including “number of letters” as covariate.

Mean accuracy scores and refixation probabilities for each subgroup and each condition are reported in Fig. 7. Analysis on accuracy scores showed no main effect of consistency and no interaction between subgroup and consistency [$F_s < 1$]. Analyses on refixation probability did not reveal either main effect of consistency [$F(1, \text{inf}) = 1.84, p = 0.18$] or interaction between subgroup and consistency [$F < 1$].

Mean fixation durations for each subgroup and condition are reported in Fig. 8. Analysis on gaze duration revealed neither main effect of consistency, nor interaction between subgroup and consistency [both $F_s < 1$]. The same pattern of results was observed when analyzing single fixation duration: no main effect of consistency [$F < 1$], and no interaction between subgroup and consistency [$F(1, 19) = 1.13, p = 0.30$].

We then examined the first and second fixation durations. Since words that were fixated more than once represent only 33.9% of the correctly recognized words on average (29.2% in sem- dyslexics, 38.7% in sem+ dyslexics), these results must be taken with caution. The analysis of first fixation duration showed no main effect of consistency [$F(1, 23) = 1.27, p = 0.27$] and no interaction between subgroup and consistency [$F(1, 16) = 1.90, p = 0.19$]. For second fixation duration, the analyses revealed a marginally significant main effect of consistency [$F(1, 18) = 3.40, p = 0.08$] but no interaction between subgroup and consistency [$F(1, 15) = 1.72, p = 0.21$].

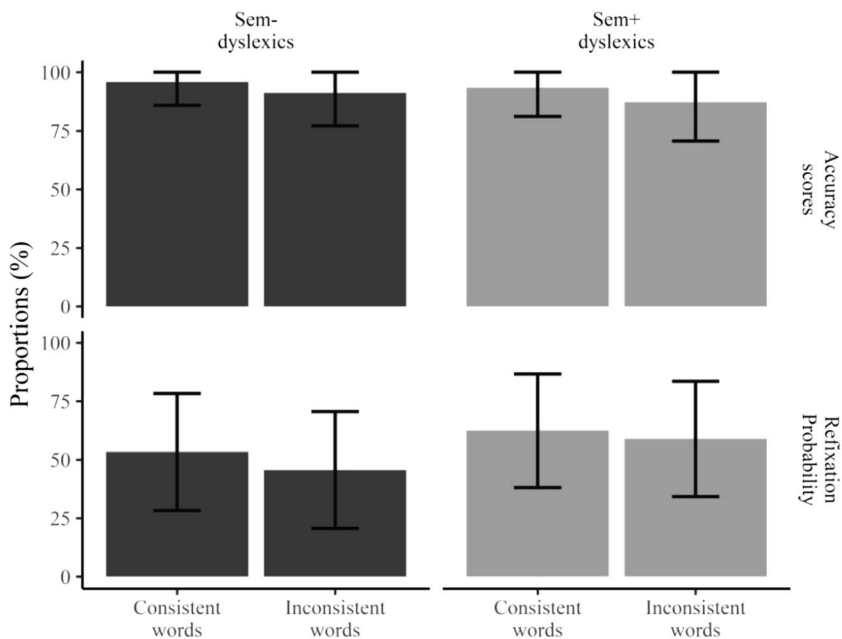


Fig. 7 Mean proportions of correct responses (top panel) and refixation probability (bottom panel) on consistent and inconsistent words in sem- (dark gray) and sem+ dyslexics (light gray). Error bars represent standard deviations

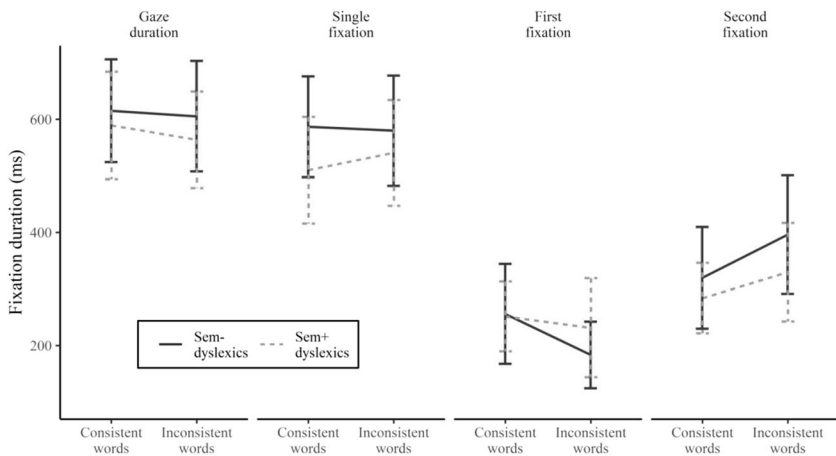


Fig. 8 Mean fixation durations on consistent and inconsistent words in sem- (dark lines) and semantic dyslexics (dashed light gray lines) for gaze duration, single, first, and second fixations. Error bars represent standard deviations

Discussion

The present study investigated task performance and eye gaze patterns of adults with and without dyslexia when they recognized visual words, as well as the contribution and the activation time-course of phonological code in this process. To this aim, we recorded the accuracy score and several coarse-grained and fine-grained parameters related to eye movements during a lexical decision task in which the phonological consistency of written words was manipulated.

Phonological skills in adults with dyslexia

Before performing the lexical decision task, all participants completed a battery of pretests that evaluated different aspects of phonological processing. The results showed that adults with dyslexia had poorer performance than skilled readers in all tasks, indicating their persistent phonological deficits in phonemic awareness (Cavalli et al., 2016; Martin et al., 2010), verbal short-term memory (Paulesu et al., 2001; Vasic, Lohr, Steinbrink, Martin, & Wolf, 2008), and pseudoword reading (Suárez-Coalla & Cuetos, 2015). The first two phonological skills, namely the ability to explicitly manipulate speech sounds and the ability to store phonological structure of words during a very short period of time, are central abilities linked to reading success and are determinant for pseudoword reading. Phonemic awareness is necessary to understand the alphabetic principle and allows the learning of letter-sound correspondences to decode written words. During the reading process, the sounds generated from the perceived letters are stored in memory, which allows a subsequent activation of the associated semantic codes. In addition, many studies suggest that during reading acquisition, the automation of this decoding procedure or phonologically mediated pathway (orth-phon-sem) allows children to develop the direct pathway (orth-sem; Windfuhr & Snowling, 2001; Ziegler, Perry, & Zorzi, 2013). Thus, the presence of deficits at the decoding stage may explain why readers with dyslexia show persistent reading deficits in adulthood.

In addition to the poor performance observed in the diagnostic tasks, eye gaze patterns recorded during the lexical decision task further contributes to a better specification of the underlying deficits in adults with dyslexia. First, we found that even in this relatively elementary task performed on familiar, high-frequency words, readers with dyslexia made more refixations and longer fixation duration than skilled readers on both global (i.e., gaze duration and single fixation duration) and fine-grained measures (i.e., first and second fixations' duration). This pattern of results obtained in French replicates those previously reported in both more and less transparent orthographic systems, such as Italian (De Luca, Borrelli, Judica, Spinelli, & Zoccolotti, 2002) and English (Rayner, 2009; Vagge, Cavanna, Traverso, & Iester, 2015; Yagle et al., 2017). It reflects a profound and general deficit which is independent of the transparency of the orthographic system (Rayner, 2009; Vagge, Cavanna, Traverso, & Iester, 2015).

Second, by combining the eye movement technique with the manipulation of word's phonological consistency, we were able to identify another aspect of the reading process that may contribute to reading difficulties, namely, the activation of phonological code. The coarsest measures (i.e., accuracy scores, gaze duration, and single fixation duration) did not show any influence of phonology during written word recognition, in either populations, which is consistent with previous studies (see also Balota, Cortese, Sergent-Marshall, Spieler, & Yap, 2004; Pexman, Lupker, & Reggin, 2002; Ziegler, Petrova, & Ferrand, 2008 for similar observations). However, the finer-grained measures (i.e., first and second fixations' durations) revealed different time-courses of phonological code activation in these two populations. Given the existence of phonological deficits in readers with dyslexia as revealed by the pretests, one could expect that the activation of the phonological code would be delayed or absent in this population compared to skilled readers whose activation of the code has been reported to be fast and automatic (Pattamadilok et al., 2017; Zeguers, Snellings, Huizenga, & van der Molen, 2014).

Indeed, the analyses of eye gaze patterns in readers with dyslexia showed that only the duration of second fixation, which reflects a late processing stage, was affected by phonological consistency. This delayed activation of the phonological code may contribute to slow written word recognition in readers with dyslexia (Boets et al., 2013; Cavalli et al., 2018). If in skilled readers, words could be recognized in parallel by both direct and phonologically mediated pathways (Castles, Bates, & Coltheart, 2006; Castles et al., 2009), the delayed activation of the phonological code in readers with dyslexia would inevitably render the phonologically mediated pathway less efficient and therefore reduce the overall speed of word recognition.

The delayed activation of the phonological code in adults with dyslexia reported here may reflect a general weakness of the reading system, which itself, was caused by inadequately developed phonological representations during oral language acquisition. As previously argued in the literature (Torppa, Lyytinen, Erskine, Eklund, & Lyytinen, 2010), a poorly developed oral language would entail poorly developed orthographic code, and diminished synchronization between the phonological and orthographic codes (Breznitz & Misra, 2003), as well as less efficient connections with the semantic code (Gentaz, Sprenger-Charolles, & Theurel, 2015).

Nonetheless, it is worth noting that more and more studies highlight the heterogeneity of the cognitive profiles in readers with dyslexia. More precisely, it has been shown that well-developed morphological (Cavalli et al., 2017; Law, Wouters, & Ghesquière, 2015) and vocabulary (Cavalli et al., 2016; Rose & Rouhani, 2012) skills are protective factors against

poor reading outcomes and can enhance both reading comprehension and reading fluency in this population (Haft, Myers, & Hoeft, 2016). Some studies also point out a discrepancy between decoding and comprehensions skills in this population (Aboud, Bailey, Petrill, & Cutting, 2016; Cutting, Bailey, Barquero, & Aboud, 2016; Patael et al., 2018; Welcome, Leonard, & Chiarello, 2010). However, the analyses comparing the influence of phonological consistency in readers with dyslexia who exhibited better vocabulary knowledge while being poor decoders (i.e., sem+ dyslexics) and those who showed the opposite profile (sem-dyslexics, who remained poor decoders when compared to skilled readers) did not show a significant interaction between consistency and dyslexics' subgroups on any measure. A marginally significant main effect of consistency was found on second fixation duration, which confirmed the result observed in the entire group of participants.

Two reasons of this absence of a significant interaction can be put forward. First, this could be due to a lack of power, as each subgroup contained only 15 individuals, which might not be enough to reveal interactions between dyslexics' subgroups and consistency, especially in this very heterogeneous population. Second, it is also possible that the task used in the present study was not the most appropriate to highlight the difference between the two subgroups since participants were presented with isolated words. Indeed, protective factors based on semantic skills may be more efficient in the presence of contextual information (Bruck, 1990; Corkett & Parrila, 2008). As mentioned in the introduction, due to their persistent and pervasive difficulties in decoding abilities, readers with dyslexia have difficulties to synchronize the phonological and orthographic codes (Breznitz & Misra, 2003), leading to less efficient connections with the semantic code (Gentaz, Sprenger-Charolles, & Theurel, 2015) and, therefore, a less automated direct pathway. As a result, even those who developed protective factors based on strong semantic skills may experience difficulties and continue to rely on inefficient connections between the orthographic and phonological codes when recognizing isolated words. In addition, all the words used in the experiment were mono-morphemic, which might not allow readers with dyslexia to take an advantage of semantic information conveyed by morphemes.

Visual word recognition in skilled readers

Taken as a group with homogeneous reading skills, our skilled reader participants did not show a phonological consistency effect neither on first nor on second fixation duration. This observation is surprising as many studies claimed that in this population, phonological code is rapidly and automatically activated during visual word recognition (Pattamadilok et al., 2017; Zeguers, Snellings, Huizenga, & van der Molen, 2014).

In an attempt to explain this unexpected finding and based on the proposal from recent studies on skilled readers that looked at the effect of phonological consistency on written word naming, we performed additional post hoc analyses. Several studies argued that the influence of phonology during reading might also depend on the balance between participants' phonological and semantic skills (Hoffman, Lambon Ralph, & Woollams, 2015; Woollams, Lambon Ralph, Madrid, & Patterson, 2016). According to these studies, skilled readers with higher phonological than semantic skills (i.e., phonological readers) seem to rely more systematically on the connections between orthographic and phonological codes when recognizing words. Reversely, readers showing higher semantic than phonological skills (i.e., semantic readers) seem to rely more systematically on the connections between orthographic and semantic codes (Woollams, Lambon Ralph, Madrid, & Patterson, 2016; Woollams & Patterson, 2012). Based

on the claim that skilled readers may not represent a homogeneous population with identical reading profile, we decided to conduct exploratory analyses of inter-individual differences in our sample of skilled readers and test the hypothesis that participants' reading profile might have affected the contribution of phonology during reading.

In other words, the literature suggests a negative correlation between semantic and phonological abilities in skilled readers. In line with this observation, we indeed found a correlation between vocabulary knowledge and phonological decoding skills which was significant in skilled readers only. This observation further justified the separation of these participants into two subgroups according to their cognitive profile.

Based on the classification proposed by Woollams, Lambon Ralph, Madrid, and Patterson (2016), skilled readers were separated into two subgroups according to their decoding skills and vocabulary knowledge. According to these authors, we expected phonological readers to be influenced by the phonological code from the earliest processing stages of visual word recognition. However, the semantic readers, who have developed strong connections between orthographic and semantic codes, would be more likely to bypass phonological code during silent reading, especially when high-frequency words are used, and thus may not show a phonological consistency effect at any processing stage.

The analyses on the global measures (i.e., gaze duration, single fixation duration, and accuracy) did not reveal any influence of the phonological consistency on written word recognition. Interestingly, the analyses of first and second fixation durations providing more fine-grained measures of the different processing stages showed that phonological readers were influenced by phonological consistency during first fixation while semantic readers did not show any phonological consistency effect. Although this exploratory finding should be considered with caution, this pattern of results suggests that the rapid activation of phonological code during reading may not be a general characteristic of skilled readers as typically assumed but may be restricted to individuals who have developed high decoding skills and consequently for whom phonological code activation is a key process when reading words.

The above observation nicely fits into the framework of the triangle reading model which assumes that the strength of the connections between orthographic, phonological, and semantic codes depends on individual's decoding and semantic skills (Harm & Seidenberg, 2004; Hoffman, Lambon Ralph, & Woollams, 2015; Woollams, Lambon Ralph, Madrid, & Patterson, 2016). Skilled readers with high decoding skills may rely more strongly on the phonological or phonologically mediated pathway (connections between orthographic and phonological codes) when reading words while skilled readers with high semantic skills may rely more strongly on the semantic or direct pathway (connections between orthographic and semantic codes). Although this model was initially created to explain the reading performance in a relatively opaque language like English where the differences between the mechanisms involved in regular and irregular word reading is discernible, our findings suggest that the same dissociation also applies to readers of a more transparent orthographic system like French.

As an initial step, the present study only focused on monosyllabic word recognition. While multisyllabic words tend to differ from monosyllabic ones in different ways (Yap & Balota, 2009), previous studies found that psycholinguistic variables like word frequency, orthographic neighborhood size, age of acquisition, imageability, or phonological consistency could similarly predict performance on both monosyllabic and multisyllabic words recognition in skilled readers (Cortese & Schock, 2013; Yap & Balota, 2009). Based on these researches, we expect that our conclusion could be generalized to multisyllabic words.

Conclusion

Using fine-grained measures provided by the eye-tracking technique, we showed that university students with dyslexia still show a delayed activation of phonological representations despite their relatively good level of vocabulary. Studying the same population, Cavalli et al. (2017) reported a spatio-temporal cortical reorganization of the reading network which reflects an earlier activation of morphological codes of words than in skilled readers. Although the present study does not allow us to draw a conclusion on a direct link between these two observations, one could speculate that high-functioning dyslexic population might have developed a reading strategy based on morphological information to compensate for their persistent phonological deficits reported here. Thus, our findings highlight the need for further research investigating the impact of phonological deficits on the different processing stages of visual word recognition and how these deficits induce a more general reorganization of the cognitive system in individuals with dyslexia.

Our results also suggest that skilled readers may not be considered a homogenous group, which may partly explain some contradictory results on consistency effects reported in the literature. The different reading profiles that we highlighted may reflect the differences in the strength of connections between the three essential codes involved in visual word recognition. Further research on the interplay between the different components of language is needed to obtain a complete picture of how language is processed in both dyslexic and heterogeneous skilled reader populations.

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Appendix

Table 6 List of the consistent and inconsistent words used in the present study and their corresponding measure of phonological consistency (Ziegler, Jacobs, & Stone, 1996)

Stimulus	Words type	Phonological consistency
Carte	Consistent	1
Chien	Consistent	1
Chiffre	Consistent	1
Cloche	Consistent	1
Culte	Consistent	1
Double	Consistent	1
Foule	Consistent	1
Fourche	Consistent	1
Gorge	Consistent	1
Groupe	Consistent	1
Guide	Consistent	1
Juge	Consistent	1
Masque	Consistent	1

Table 6 (continued)

Stimulus	Words type	Phonological consistency
Ordre	Consistent	1
Prince	Consistent	1
Sable	Consistent	1
Singe	Consistent	1
Stage	Consistent	1
Vague	Consistent	1
Forme	Consistent	1
Quille	Inconsistent	0.494
Chef	Inconsistent	0.826
Short	Inconsistent	0.017*
Clef	Inconsistent	0.174
Cap	Inconsistent	0.165
Drap	Inconsistent	0.835
Femme	Inconsistent	0.997
Fille	Inconsistent	0.494
Flemme	Inconsistent	0.003
Grille	Inconsistent	0.494
Gaz	Inconsistent	0.948
Gang	Inconsistent	0.003
Jus	Inconsistent	0.002
Mail	Inconsistent	0.106*
Os	Inconsistent	0.033
Prix	Inconsistent	0.255
Sens	Inconsistent	0.391
Stand	Inconsistent	0.001
Stop	Inconsistent	0.006
Ville	Inconsistent	0.507

*The degree of phonological consistency was computed from the [Lexique.org](https://www.lexique.org/) database (New, Pallier, Ferrand, & Matos, 2001) following the same criterion and methodology as described by Ziegler, Jacobs, and Stone (1996)

Table 7 Regression coefficients of the main effects of subgroup, consistency, and their interaction (for each dependent variable separately) calculated through LME/gLME models when analyzing the inter-individual differences in skilled readers (** $p < 0.01$; * $p < 0.05$; ns: $p > 0.10$). Models: dependent variable ~ SR subgroup $\times$ consistency + number of letters + (1 + consistency | subject) + (1 + SR subgroup | item)

Dependent variable		Degrees of freedom	<i>F</i> ratio	<i>p</i> values
Accuracy scores	Subgroup	1, inf	2.53	ns
	Consistency	1, inf	0.21	ns
	Subgroups $\times$ consistency	1, inf	0.12	ns
Gaze duration	Subgroup	1, 26	4.39	*
	Consistency	1, 27	0.95	ns
	Subgroups $\times$ consistency	1, 21	1.21	ns
Single fixation duration	Subgroup	1, 26	5.14	*
	Consistency	1, 26	0.26	ns
	Subgroups $\times$ consistency	1, 21	1.68	ns
First fixation duration	Subgroup	1, 21	0.16	ns
	Consistency	1, 19	0.003	ns
	Subgroups $\times$ consistency	1, 9	6.13	*
Second fixation duration	Subgroup	1, 21	1.40	ns
	Consistency	1, 17	0.38	ns
	Subgroups $\times$ consistency	1, 11	0.25	ns
Refixation probability	Subgroup	1, inf	10.17	**
	Consistency	1, inf	0.00	ns
	Subgroups $\times$ consistency	1, inf	0.80	ns

Table 8 Regression coefficients of the main effects of subgroup, consistency, and their interaction (for each dependent variable separately) calculated through LME/gLME models when analyzing the inter-individual differences in readers with dyslexia (–: $p < 0.075$; ns: $p > 0.10$). Models: dependent variable ~ DYS subgroup × consistency + orthographic uniqueness point + (1 + consistency | subject) + (1 + DYS subgroup | item)

Dependent variable		Degrees of freedom	F ratio	p values
Accuracy scores	Subgroup	1, inf	0.70	ns
	Consistency	1, inf	1.13	ns
	Subgroups × consistency	1, inf	0.03	ns
Gaze duration	Subgroup	1, 31	0.69	ns
	Consistency	1, 43	0.38	ns
	Subgroups × consistency	1, 36	0.02	ns
Single fixation duration	Subgroup	1, 29	2.69	ns
	Consistency	1, 29	0.51	ns
	Subgroups × consistency	1, 20	1.11	ns
First fixation duration	Subgroup	1, 20	0.79	ns
	Consistency	1, 23	1.60	ns
	Subgroups × consistency	1, 16	1.81	ns
Second fixation duration	Subgroup	1, 24	1.74	ns
	Consistency	1, 17	3.61	–
	Subgroups × consistency	1, 15	1.70	ns
Refixation probability	Subgroup	1, inf	2.36	ns
	Consistency	1, inf	1.13	ns
	Subgroups × consistency	1, inf	0.60	ns

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